MTH2010

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

January 2021

Groups Rings and Fields

Module Leader: Mohamed Saïdi

Duration: 2 HOURS (+30 minutes upload time)

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) (i) Find a cyclic normal subgroup H of D_8 with cardinality 4. (8)

   (ii) Let H be a cyclic normal subgroup of D_8 with cardinality 4 as in (i). By constructing a surjective homomorphism $\phi : D_8 \to \mathbb{Z}/2\mathbb{Z}$ with kernel H; or otherwise, prove that D_8/H is isomorphic to $\mathbb{Z}/2\mathbb{Z}$. (8)

   (b) Let p be a prime number and G a finite group with p elements. Prove that G is a cyclic group. (8)

   (c) (i) Find the Sylow 5-subgroups of D_{10}. (6)

   (ii) Find two Sylow 3-subgroups H_1 and H_2 of the alternating group A_4 and an element $\sigma \in A_4$ such that $H_2 = \sigma H_1 \sigma^{-1}$. (6)

   (d) Let R be the ring

$$R = \mathbb{Z}[i] = \{a + ib : a, b \in \mathbb{Z}\}$$

   and $N : R \to \mathbb{Z}$ the norm on R defined by

$$N(a + ib) = a^2 + b^2.$$

   (i) Show that an element u of R is a unit if and only if $N(u) = 1$. You may use the fact that $N(xy) = N(x)N(y)$ for $x, y \in R$. (4)

   (ii) Find the units of R and show that the group $R^\times$ of units of R is a cyclic group. (6)

   (iii) Find a cyclic subgroup of $R^\times$ of cardinality 2. (4)

[**50**]

SECTION B

2. Let G be a finite group.

   (a) Define the centre $Z(G)$ of G and for an element $g \in G$ define the centraliser $C_G(g)$ of g in G. Given $g \in G$, prove that g does NOT belong to $Z(G)$ if and only if the index $|G : C_g(g)|$ of $C_G(g)$ in G is bigger than 1. (10)

   (b) Let $g \in G$ and
   $$\text{orb}(g) = \{hgh^{-1} : h \in G\}$$
   the orbit of g under conjugation. Let $n = |G : Z(G)|$ be the index of $Z(G)$ in G. Prove that the cardinality $|\text{orb}(g)|$ of $\text{orb}(g)$ satisfies $|\text{orb}(g)| \leq n$. (8)

   (c) Write down the class equation for G. Assuming $|G| = p^n$, with $n \geq 1$ and p is a prime number, prove that $|Z(G)| > 1$. (7)

   [25]

3. (a) Let $n \geq 1$ be an integer and p an odd prime number dividing $2n$. Show that a Sylow p-subgroup of D_{2n} is cyclic and normal. (12)

   (b) Let
   $$R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$$
   and
   $$I = (3, 2 + \sqrt{-5})_R$$
   the ideal of R generated by 3 and $2 + \sqrt{-5}$.

   By constructing a surjective homomorphism $\phi : R \to K$ from R to a field K with kernel $\text{Ker}(\phi) = I$, or otherwise, prove that I is a maximal ideal of R. (13)

   [25]

4. (a) Let R be a ring. Define what it means for an ideal of R to be a prime ideal. Let P be a prime ideal of R and I, J ideals of R such that $I \cap J$ is contained in P. Prove that either $I \subset P$ or $J \subset P$. (10)

(b) Let R be the ring

$$R = \mathbb{Z}[\sqrt{-11}] = \{a + b\sqrt{-11} : a, b \in \mathbb{Z}\}$$

and $N : R \to \mathbb{Z}$ the norm on R defined by

$$N(a + b\sqrt{-11}) = a^2 + 11b^2.$$

(i) Prove that the elements 2, $1 + \sqrt{-11}$, and $1 - \sqrt{-11}$ of R are irreducible. You may use the facts that an element $u \in R$ is a unit if and only if $N(u) = 1$, and $N(xy) = N(x)N(y)$ if $x, y \in R$. (9)

(ii) Prove that R is NOT a unique factorisation domain. (6)

[**25**]